Quadratic functions and technology

1 Graph each function using a calculator or other technology then answer the following questions:

i State the coordinates of the vertex.

ii State the coordinates of the y-intercept.

a $y = 4x^2 - 64x + 263$

b $y = 4x^2 + 48x + 135$

c $y = -4x^2 - 48x - 140$

d $y = -3x^2 + 12x - 13$

e $y = 3x^2 + 3x + 9$

f $y = 4x^2 + 3x - 1$

2 Graph each function using a calculator or other technology then answer the following questions:

i State the coordinates of the vertex of the graph.

ii Are there any x-intercepts? If yes, find the x-value(s) of the x-intercept(s).

iii For $x > 0$, is the parabola increasing or decreasing?

a $y = 4x^2 + 16$

b $y = -3x^2 - 12$

c $y = -5x^2 + 125$

d $y = 4x^2 - 64$

3 State whether the following parabolas have x-intercepts:

a $y = (x - 7)^2 + 4$

b $y = (x - 7)^2 - 4$

c $y = -(x - 7)^2 + 4$

d $y = -(x - 7)^2 - 4$

4 Find the value of c if the parabola $y = x^2 + 10x + c$ has exactly one x-intercept.

5 Graph each function using a graphing calculator then answer the following questions:

i Find the x-value(s) of the x-intercept(s).

ii Find the y-value of the y-intercept.

iii State the coordinates of the vertex.

a $y = \left(x + \dfrac{5}{2}\right)^2$

b $y = -(x - 3)^2 + 4$

c $y = 9 - (x - 4)^2$

6 Consider the parabolas with equations $f(x)$ [illegible] $+ 2x + 4$ and $g(x) = x^2 - 2x + 4$:

a Using a graphing calculator, state the coordinates of the vertex of each parabola:

i For $f(x)$

ii For $g(x)$

b How far apart are the vertices of the two parabolas?

7 Will the curve $y = -x^2 - 5x + 1$ reach a maximum or a minimum value as it turns?

8 Graph $y = x^2 - 4$ using a calculator or other technology then answer the following questions:

a Does it have a minimum or maximum y-value?

b State the coordinates of the vertex.

9 Using a graphing calculator or other technology, find the minimum value of y for the quadratic function $y = x^2 + x - 20$.

10 Graph $y = x^2 + 6.2x - 7$ using a caculator or any technology then answer the following questions:

a State the equation of the axis of symmetry.

b Find the minimum value of y.

11 Graph the following functions using a graphing calculator then answer the following questions:

i Find the x-value(s) of the x-intercept(s).

ii Find the y-value of the y-intercept.

iii State the equation of the axis of symmetry.

iv State the coordinates of the vertex.

a $y = x(6 - x)$

b $y = x^2 + 2x$

c $y = -x^2 - 4x$

d $y = (x + 1)(x - 3)$

e $y = x^2 + 2x - 3$

f $y = -2x^2 + 16x - 24$

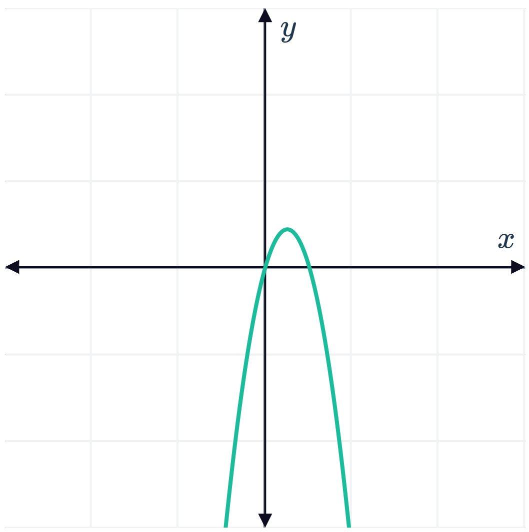
12 Consider the function $y = -0.68x^2 + 3.48x$:

a. Find the x-coordinate of the vertex, correct to two decimal places.

b. Find the y-coordinate of the vertex, correct to two decimal places.

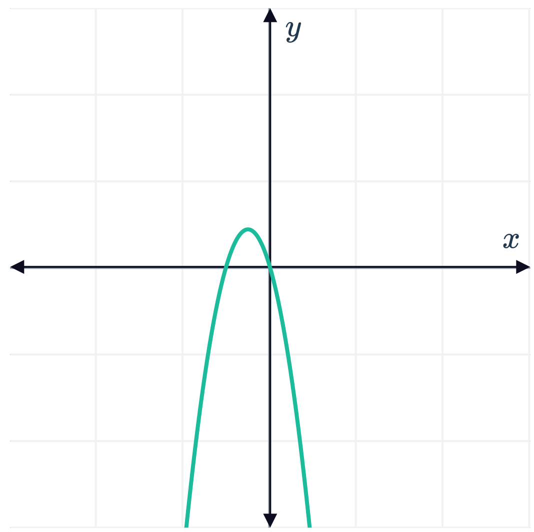
c. State whether the following are possible viewing window on your calculator that shows the vertex and the x-intercepts:

i



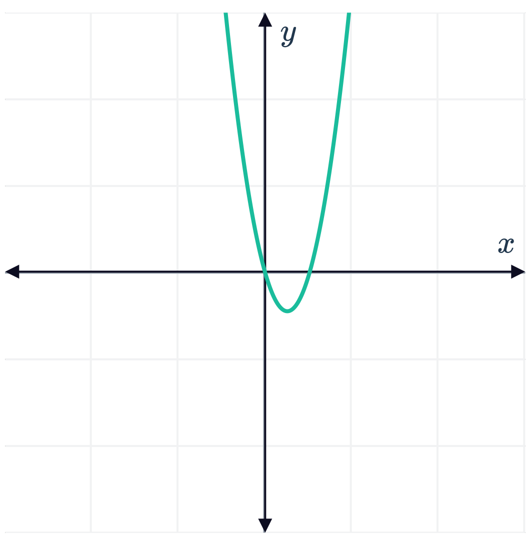
$-20, 30, 5$ by $-20, 30, 5$

ii



$-20, 30, 5$ by $-20, 30, 5$

iii



$-20, 30, 5$ by $-20, 30, 5$

d. Use a graphing calculator to find the x-intercepts, correct to two decimal places.

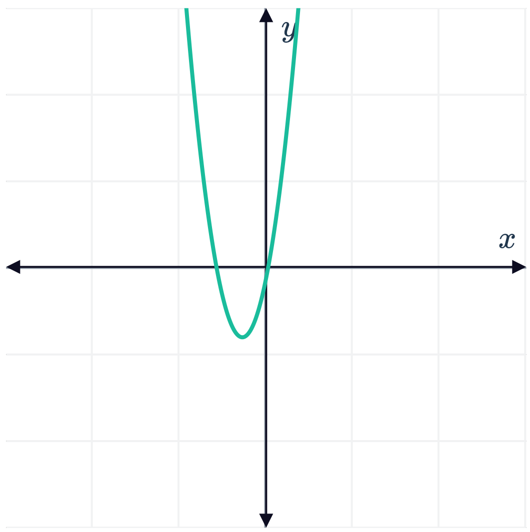
13 Consider the function $y = 0.91x^2 - 5x - \sqrt{5}$

a Find the x-coordinate of the vertex, correct to two decimal places.

b Find the y-coordinate of the vertex, correct to two decimal places.

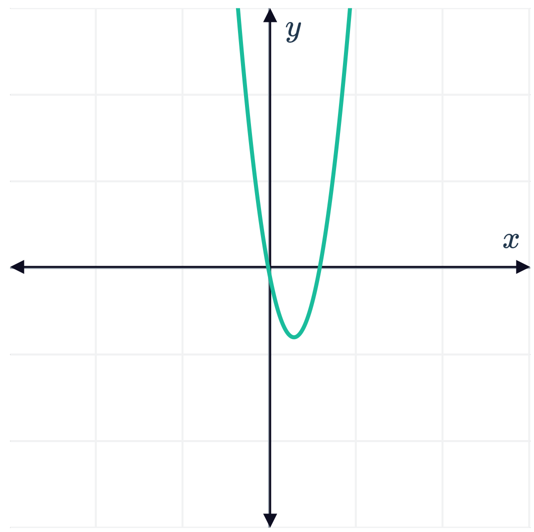
c State whether the following are possible viewing window on your calculator that shows the vertex and the x-intercepts:

i



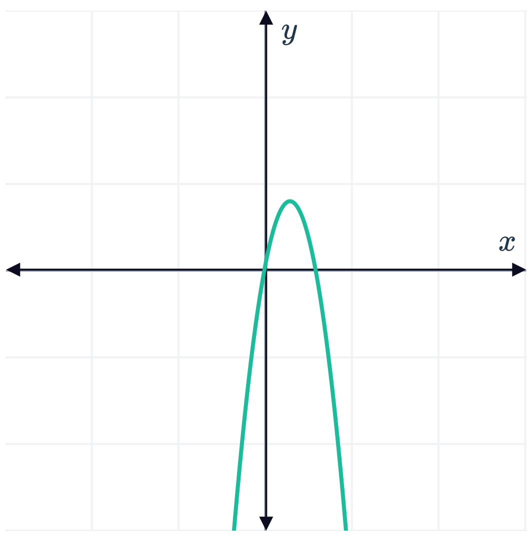
$-20, 30, 5$ by $-30, 20, 5$

ii



$-20, 30, 5$ by $-30, 20, 5$

iii



$-20, 30, 5$ by $-30, 20, 5$

d Use a graphing calculator to find the x-intercepts to two decimal places.

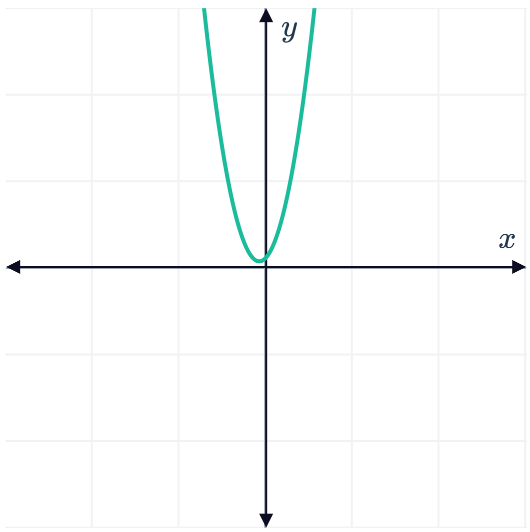
14 Consider the function $y = -0.72x^2 + \sqrt{5}x +$ [redacted]

a Find the x-coordinate of the vertex, correct to two decimal places.

b Find the y-coordinate of the vertex, correct to two decimal places.

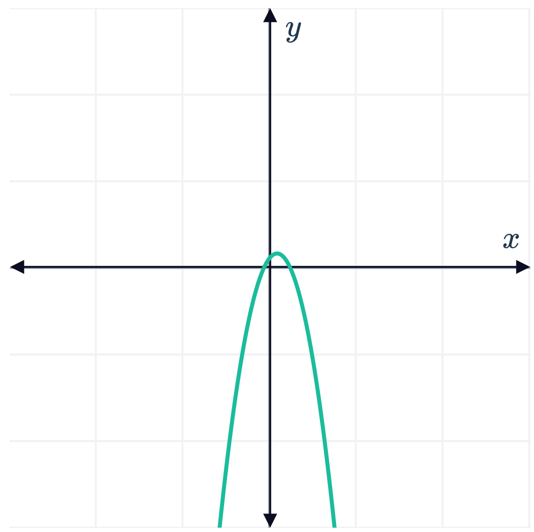
c State whether the following are possible viewing window on your calculator that shows the vertex and the x-intercepts:

i



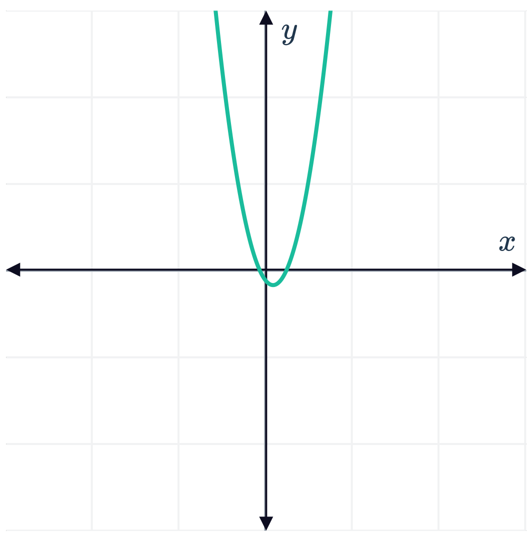
$-20, 30, 5$ by $-20, 30, 5$

ii



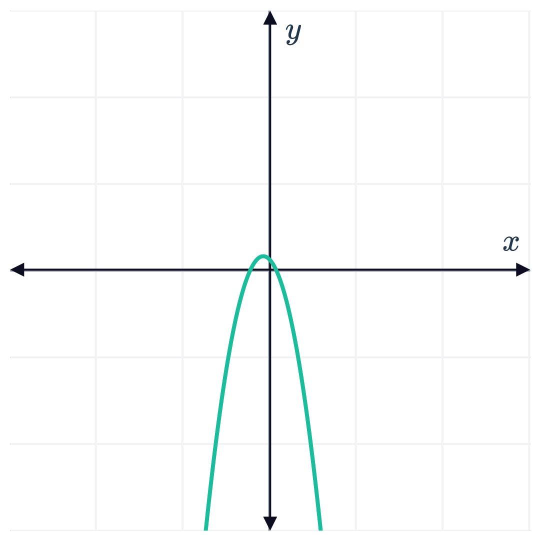
$-20, 30, 5$ by $0, 30, 5$

iii



$-20, 30, 5$ by $-20, 30, 5$

iv



$-20, 30, 5$ by $-20, 30, 5$

d Use a graphing calculator to find the x-intercepts, correct to two decimal places.

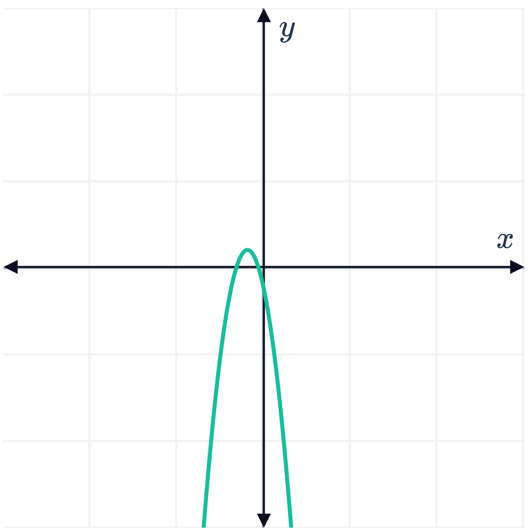
15 Consider the function $y = -\sqrt{3}x^2 - 4.79x -$ [redacted]

a Find the x-coordinate of the vertex, correct to two decimal places.

b Find the y-coordinate of the vertex, correct to two decimal places.

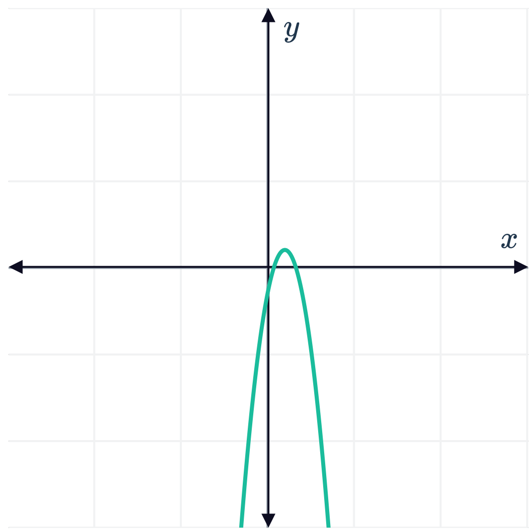
c State whether the following are possible viewing window on your calculator that shows the vertex and the x-intercepts:

i



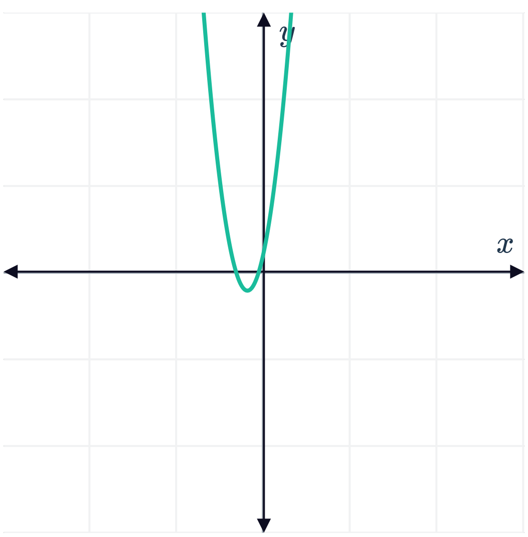
$-30, 20, 5$ by $-20, 30, 5$

ii



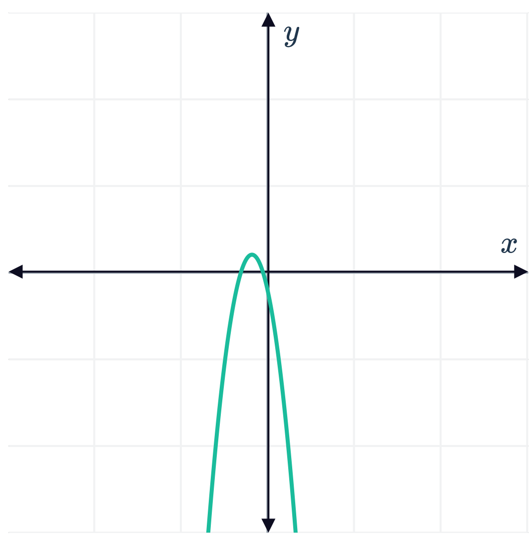
$-30, 20, 5$ by $-20, 30, 5$

iii



$-30, 20, 5$ by $-20, 30, 5$

iv



$-30, 20, 5$ by $0, 30, 5$

d Use a graphing calculator to find the x-intercepts, correct to two decimal places.

16 How do we shift the graph of $y = x^2$ to get the graph of $y = x^2 - 5$?

17 How do we shift the graph of $y = x^2$ to get the graph of $y = (x-3)^2$?

18 Using a calculator or other technlogy, graph [redacted]unctions $y = x^2$, $y = 4x^2$, and $y = 6x^2$. For $y = ax^2$, as a increases, how does it change the graph of $y = x^2$?

19 Using a graphing calculator, graph $y = x^2$ and write the equation needed for the following translations:

a $y = x^2$ shifted 2 units up

b $y = x^2$ reflected about the x-axis

c $y = x^2$ shifted 2 units up then reflected about the x-axis

d $y = x^2$ is translated 2 units to the left and 2 units up

20 A parabola of the form $y = (x - h)^2 + k$ is symmetrical about the line $x = 2$, and its vertex lies 3 units above the x- axis. Write the equation needed to graph the parabola into a graphing calculator.

21 Solve each of the following equations by splitting the equation into two functions and graphing them using a calculator or other technology:

a $7x^2 = 8x$

b $2x\left(x - \dfrac{5}{2}\right) = 3$

c $2x\left(x - 1\right) = -x + 3$

d $\dfrac{5 + 19x}{4x} = x$

e $\left(x + 1\right)\left(x - 2\right) = -3x - 3$

f $\dfrac{x}{x+2} + \dfrac{9}{x-2} = \dfrac{6}{x^2 - 4}$

22 Graph each function using a calculator or other technology then determine the number of unique solutions to the corresponding equation:

a Function: $f\left(x\right) = \left(x - 9\right)^2$
Equation: $\left(x - 9\right)^2 = 0$

b Function: $f\left(x\right) = \left(x - 3\right)\left(x - 2\right)$
Equation: $\left(x - 3\right)\left(x - 2\right) = 0$

c Function: $f\left(x\right) = -x^2 + 8x - 17$
Equation: $-x^2 + 8x - 17 = 0$

23 Find the solution(s) of $3x^2 - 11x + 10 = 0$, correct to three decimal places.

24 Find the zero(s) of $f\left(x\right) = 3.83x^2 - 2.78x - 5.21$, correct to three decimal places.

Additional questions

25 The Millau Viaduct in Millau, Southern France, is $270 \mathrm{~m}$ tall. We can model the behaviour of objects falling from the viaduct using Galileo's formula for falling objects: $d=4.9 t^{2}$, where d is distance fallen in meters and t is time in seconds since the object was dropped.
Graph this equation using a calculator or other technology then answer the following questions:

a State the coordinates of the vertex of the graph.

b State what the answer in part (a) represent.

26 A rectangle is to be constructed with $60 \mathrm{~m}$ of wire. The rectangle will have an area of $A=30 x-x^{2}$, where x is the length of one side of the rectangle.
Graph this equation using a calculator or other technology then answer the following questions:

a Find the x-values of the x-intercepts.

b Find the x-value of the vertex.

c State what the height of the parabola at $x=15$ represent.

27 The kinetic energy E of a moving object is given by $E=\frac{1}{2} m v^{2}$, where m is its mass in kilograms and v is its speed in meters/second.
Graph this equation for a vehicle with a mass of $1400 \mathrm{~kg}$ using a calculator or other technology. Then, calculate the velocity when the kinetic energy is $137\,200 \mathrm{~J}$.

28 An object launched from the ground has a height (in meters) after t seconds that is modelled by the equation $y=-4.9 t^{2}+58.8 t$.
Graph this equation using a calculator or other technology then answer the following questions:

a Find the maximum height of the object.

b After how many seconds is the object at its maximum height?

c After how many seconds does the object return to the ground?

29 An object is released $900\ \mathrm{m}$ above ground and falls freely. The distance the object is from the ground is modelled by the formula $d = 900 - 4.9t^2$, where d is the distance in meters that the object falls and t is the time elapsed in seconds.
Graph this equation using a calculator or other technology then answer the following questions:

a Find d when $t = 0$.

b Explain what the answer in part (a) mean.

c Find the value of t when $d = 0$, correct to two decimal places.

d Explain what the answer in part (c) mean.

30 On the moon, the equation $d = 0.8t^2$ is used to find the distance an object has fallen after t seconds. On Earth, the equation is $d = 4.9t^2$.

a A rock is thrown from a height of $80\ \mathrm{m}$ on Earth. Graph the equation on a calculator or other technology to find the time taken to hit the ground to the nearest second.

b A rock is thrown from a height of $80\ \mathrm{m}$ on the Moon. Graph the equation on a calculator or other technology to find the time taken to hit the ground to the nearest second.